

Hepatoma tissue culture (HTC) cells as a model for investigating the effects of low concentrations of herbicide on cell structure and function

M. Malatesta^{a,*}, F. Perdoni^b, G. Santin^b, S. Battistelli^c, S. Muller^d, M. Biggiogera^{b,e}

^a Dipartimento di Scienze Morfologico-Biomediche, Sezione di Anatomia e Istologia, University of Verona, Strada Le Grazie, 8, 37134 Verona, Italy

^b Laboratorio di Biologia Cellulare e Neurobiologia, Dipartimento di Biologia Animale, University of Pavia, Pavia, Italy

^c Istituto di Istologia e Analisi di Laboratorio, University of Urbino, Urbino (PU), Italy

^d CNRS, Institut de Biologie Moléculaire et Cellulaire, Immunologie et Chimie Thérapeutiques, Strasbourg, France

^e Istituto di Genetica Molecolare del CNR (Sezione di Istochimica e Citometria), Pavia, Italy

ARTICLE INFO

Article history:

Received 7 July 2008

Accepted 8 September 2008

Available online 18 September 2008

Keywords:

Cell nucleus

Cytometry

Electron microscopy

Fluorescence microscopy

Herbicide

HTC cells

Lysosomes

Mitochondria

ABSTRACT

Previous studies on mice fed genetically modified (GM) soybean demonstrated modifications of the mitochondrial functions and of the transcription/splicing pathways in hepatocytes. The cause(s) of these alterations could not be conclusively established but, since the GM soybean used is tolerant to glyphosate and was treated with the glyphosate-containing herbicide Roundup™, the possibility exists that the effects observed may be due to herbicide residues. In order to verify this hypothesis, we treated HTC cells with 1–10 mM Roundup and analysed cellular features by flow cytometry, fluorescence and electron microscopy.

Under these experimental conditions, the death rate and the general morphology of HTC cells were not affected, as well as most of the cytoplasmic organelles. However, in HTC-treated cells, lysosome density increased and mitochondrial membranes modified indicating a decline in the respiratory activity. Moreover, nuclei underwent morpho-functional modifications suggestive of a decreased transcriptional/splicing activity.

Although we cannot exclude that other factors than the presence of the herbicide residues could be responsible for the cellular modifications described in GM-fed mice, the concordance of the effects induced by low concentrations of Roundup on HTC cells suggests that the presence of Roundup residues could be one of the factors interfering with multiple metabolic pathways.

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1. Introduction

Genetically modified (GM) crops, in which new genes have been inserted into the original genome, are nowadays distributed all over the world (Sanvido et al., 2007). Although no direct evidence has been reported so far that GM food may affect human or animal health, the debate is still open (e.g. Paoletti and Pimentel, 2000; Kuiper et al., 2004). In particular, studies on young, adult (2–8 months of age) and old (24 months of age) mice fed GM soybean have demonstrated hepatocyte nuclei modifications involving structural constituents of the transcription/splicing pathways as well as morpho-functional changes in mitochondria (Malatesta et al., 2002a, 2008). The cause(s) of the observed alterations could

not be conclusively established but it is worth noting that these modifications disappeared when GM soybean was replaced by a non-GM one in the diet (Malatesta et al., 2005).

The GM soybean used in these studies is tolerant to glyphosate (Padgett et al., 1995) and the plants were consequently treated in the field with the glyphosate-containing herbicide Roundup™; therefore, the possibility exists that the effects observed may be due, at least in part, to herbicide residues. It must be noted that Roundup is almost immediately degraded in the soil, where it gives rise to different metabolites, which not only persist therein (e.g. see Mamy et al., 2008) but that can also be found in the crops (e.g. see Arregui et al., 2004). Unfortunately, the very rapid degradation of Roundup and the heterogeneity of its degradation products make quite problematic their detection even in the crop; consistently, their final amounts in a standard fodder can be under the technical detection limit.

Roundup at high concentrations has been shown to deregulate cell cycle and inhibit transcription (Marc et al., 2002, 2004, 2005), alter cytoskeletal organization (Ovidi et al., 2001), interfere with steroid hormone biosynthesis (Walsh et al., 2000; Richard

Abbreviations: DFC, dense fibrillar component; EDTA, ethylene diamino tetraacetic acid; FC, fibrillar centre; GC, granular component; GM, genetically modified; HTC, hepatoma tissue culture; NGS, normal goat serum; PBS, phosphate-buffered saline; PG, perichromatin granule.

* Corresponding author. Tel.: +39 045 8027155; fax: +39 045 8027163.

E-mail address: manuela.malatesta@univr.it (M. Malatesta).

et al., 2005), depress mitochondrial respiratory activity (Peixoto, 2005) and modify liver enzymatic activity (e.g. Hietanen et al., 1983; Jiraungkoorskul et al., 2003).

In an attempt to clarify the potential role of Roundup residues in the occurrence of the cellular modifications found in mice fed glyphosate-tolerant GM soybean, we carried out some experiments on an *in vitro* cell model, which represents a simple, sensitive and reliable system to evaluate pesticide toxicity (El-Demerdash et al., 2001; Bertheussen et al., 1997), provided that its intrinsic limitations in reproducing the complex responsiveness of an organism are taken in the proper account.

Rat hepatoma tissue culture (HTC) cells were incubated with very low concentrations of Roundup for different time periods, with the aim of mimicking the exposure to herbicide traces or residues, and some cellular features were analysed by combining flow cytometry, fluorescence and electron microscopy. The structural and functional modifications induced *in vitro* have been compared with those previously observed *in vivo* on the hepatocytes from mice fed GM soybean.

2. Materials and methods

2.1. Cell culture and treatments

HTC cells (ATCC, Rockville, MD) were grown in Dulbecco's minimal essential medium supplemented with 10% (v/v) fetal calf serum, 1% (w/v) glutamine, 100 U of penicillin and streptomycin (Celbio, Milan, Italy), in a 5% CO₂ humidified atmosphere. Cells were trypsinized when subconfluent and seeded on glass coverslips in multiwell dishes for fluorescence microscopy, or in 25 cm² plastic Petri dishes for electron microscopy and flow cytometry.

Twenty-four hours later, cells were treated for 4, 12, 24 or 48 h with Roundup (Monsanto Europe N.V., Antwerp, Belgium), which was dissolved in complete medium at a concentration ranging from 1 to 10 mM. These concentrations were chosen since they are considerably lower than those previously used in the literature (e.g. Peixoto, 2005). The experiments were carried out in triplicate. Whenever not specified, the chemicals used were from Sigma (Buchs, Switzerland).

2.2. Cell viability assay

Viability was assessed by staining both control and Roundup-treated cells with Trypan blue (0.1% in the medium for 2 min). Cells that were permeable to Trypan blue were counted with a hemocytometer and considered as non-viable. Data were expressed as the mean of three independent experiments $\pm$ standard deviation.

2.3. Cytometric evaluation of the DNA content

Cells were harvested by mild trypsinization (0.25% (w/v) trypsin containing 0.05% (w/v) ethylene diamino tetraacetic acid (EDTA) in phosphate-buffered saline (PBS)) and stained for 30 min with the DNA-specific dye, propidium iodide (PI; 50 μ g/ml in PBS containing 100 U/ml of RNase A, 10 μ M EDTA, and 0.015% (v/v) Nonidet P40). To estimate the distribution of cells in the different cycle phases, cytometric measurements were taken with a Partec PAS III cytometer (Partec GmbH, Görlitz, Germany) equipped with an argon ion laser (20 mW output power) at 488 nm, with a 610 nm longpass filter for the PI red fluorescence detection. At least 20,000 cells per sample were measured. The data were handled using FlowMax software from the same company.

2.4. Cell morphometry

Phase contrast micrographs were taken with a Camedia 5050 digital camera mounted on an Olympys BX51 microscope (Olympus Italia Srl, Milan, Italy). The cellular and nuclear areas of 60 cells per sample were measured using the Image J (NIH) software, and the nucleus to cytoplasm (N/C) ratio was calculated.

2.5. Organelle immunolabelling

Different immunolabelling procedures for fluorescence microscopy were used to detect microtubules, mitochondria, the Golgi apparatus or lysosomes in cells grown on coverslips and fixed with 4% (v/v) formaldehyde (30 min at room temperature) and 70% (v/v) ethanol in water (30 min at -20° C). After rehydration with PBS, the samples were incubated with the primary and secondary antibodies reported in Table 1. Microfilaments were also labelled for 60 min at room temperature with an Alexa594-conjugated phalloidin (1:40 in PBS) (Molecular Probes, Invitrogen, Italy). In some experiments, dual-immunolabellings were also performed. All the immunolabelled samples were finally stained for DNA with Hoechst 33258 (0.1 μ g/ml in PBS for 10 min), rinsed in PBS, and mounted in Mowiol (Calbiochem, Inalco, Milan, Italy).

To estimate the density of lysosomes in the cytoplasm (number of lysosomes/100 μ m²), micrographs of the immunolabelled cell samples were taken with an Olympus BX51 microscope (see below) and the area of 80 cells from controls or Roundup-treated cells was measured using the software Image J (NIH).

2.6. Conventional and confocal fluorescence microscopy

An Olympus BX51 microscope equipped with a 100W mercury lamp (Olympus Italia Srl, Milan, Italy) was used under the following conditions: 450–480 nm excitation filter (excf), 500 nm dichroic mirror (dm), and 515 nm barrier filter (bf) for Alexa 488 and for JC-1; 540 nm excf, 580 nm dm, and 620 nm bf for Alexa 594; 330–385 nm excf, 400 nm dm, and 420 nm bf, for Hoechst 33258. Images were also recorded with an Olympus Magnifire digital camera system (Olympus Italia Srl, Milan, Italy), and stored on a PC by the Olympus software, for processing and printing.

For confocal laser scanning microscopy, we used a Leica TCS-SP system mounted on a Leica DMIRBE inverted microscope (Leica Microsystems Italia, Milan, Italy); for fluorescence excitation, an Ar/UV laser at 364 nm was used for Hoechst 33258, an Ar/Vis laser at 488 nm for FITC and an He/Ne laser at 543 nm for Alexa 594. Spaced (0.5 μ m) optical sections were recorded using a 63 $\times$ oil immersion objective. Images were collected in the 1024 $\times$ 1024 pixel format, stored on a magnetic mass memory and processed by the Leica confocal software.

Table 1
Antibodies used in the immunolabelling experiments

Antibody against	Source	Dilution in PBS	Secondary marker
Alpha-tubulin	Mouse, monoclonal (cell signaling Technology, Danvers, MA, USA)	1:500	Anti-mouse IgG
Mitochondria	Human, autoimmune serum	1:150	Anti-human IgG
Golgi apparatus	Human, autoimmune serum	1:250	Anti-human IgG
Lysosomes	Human, autoimmune serum	1:500	Anti-human IgG

The cells were incubated, at room temperature, for 1–2 h with the primary antibody and for 1 h with the secondary antibodies diluted 1:200 in PBS; the latter were conjugated with either Alexa 488 or Alexa 594.

2.7. Cytometric detection of the changes in the mitochondrial membrane potential

The changes in the mitochondrial membrane potential were monitored using JC-1 (5,5V,6,6V-tetrachloro-1,1V,3,3V-tetraethylbenzimidazolcarbocyanine iodide) (Invitrogen, Carlsbad, CA, USA). When administered to living cells, JC-1 accumulates in the mitochondria where it emits either red or green fluorescence, depending on the mitochondrial membrane potential, the red signal indicating polarized mitochondria, and the green signal the depolarized ones (Reers et al., 1991); therefore, the shift from red to green fluorescence is considered a reliable indication of a drop in the mitochondrial membrane potential. HTC cells grown in flasks were harvested by mild trypsinisation as above, and incubated in culture medium with 2 μ M JC-1 for 20 min at 37 °C in the dark. After two washes with PBS at 37 °C, the cells in suspension were analysed on a Flow cytometer Partec PAS III (Partec GmbH, Görlitz, Germany) equipped with an argon ion laser with 20 mW output power at 488 nm excitation with 530/30 nm and 585/42 nm bandpass emission filters. Data were analysed using FlowMax software from the same company.

2.8. Transmission electron microscopy

For conventional morphology, cells were fixed with 2.5% (v/v) glutaraldehyde and 2% (v/v) paraformaldehyde in 0.1 M Sörensen phosphate buffer, pH 7.4, at 4 °C for 1 h, washed, post-fixed with 1% OsO₄ at 4 °C for 30 min, dehydrated with acetone and embedded in Epon. For immunoelectron microscopy, cells were fixed with 4% paraformaldehyde in 0.1 M Sörensen phosphate buffer at 4 °C for 1 h, washed, treated with NH₄Cl 0.5 M in PBS, dehydrated with ethanol and embedded in LRWhite resin.

Ultrathin sections from Epon-embedded samples were conventionally stained with uranyl acetate and lead citrate, while LRWhite-embedded samples were stained with the EDTA technique (Bernhard, 1969) to visualize the ribonucleoprotein (RNP) components.

All samples were observed in a Philips Morgagni transmission electron microscope (FEI Company Italia Srl, Milan, Italy) operating at 80 kV and equipped with a Megaview III camera (FEI Company Italia Srl, Milan, Italy) for digital image acquisition.

Morphometrical analyses were carried out on control cells and on samples treated with 10 mM Roundup for 4 and 24 h, by using the AnalySIS Image processing software (Soft Imaging System GmbH, Germany). In Epon-embedded samples the sectional area as well as inner and outer membrane profile length were measured ($\times 26,000$) in 30 mitochondria per cell sample, then the inner/outer membrane length ratio was calculated in order to estimate the inner membrane length independently from mitochondrial size. In LRWhite-embedded samples measurements were made on 15 cell nuclei per sample ($\times 26,000$): nucleolar areas, fibrillar centres (FC) area, percentages of FC, dense fibrillar component (DFC) and granular component (GC) per nucleolus, perichromatin granule density (PG number/ μ m² of nucleoplasm) and nuclear pore frequency (number/ μ m of perimeter) were considered.

Ultrathin sections from LRWhite-embedded samples were processed also for immunocytochemistry using polyclonal antibodies directed against DNA/RNA hybrid molecules (Testillano et al., 1994), specifically occurring in the transcriptional sites. These antibodies were raised by using synthetic polyApolydT hybrid co-administrated intramuscularly in a rabbit in the presence of methylated bovine serum albumin (w/w) and Freund's adjuvant (v/v). These antibodies did not react in ELISA with dsDNA from different sources or with polyAdT. Sections, placed on nickel grids coated with a formvar-carbon layer, were floated on normal goat serum (NGS) diluted 1:100 in PBS and incubated for 17 h at 4 °C with

the primary antibody diluted in a solution containing 0.1% (w/v) bovine serum albumin and 0.05% (v/v) Tween 20 in PBS. After rinsing, sections were floated in NGS and then reacted for 20 min with the specific gold-conjugated secondary antibody (Jackson Immuno Research Laboratories Inc., West Grove, PA, USA) diluted 1:10 in PBS. Finally, sections were rinsed and air dried. As controls, some grids were treated with the incubation mixture without the primary antibody and then processed as described above. The immunolabelled samples were finally stained with the EDTA technique (Bernhard, 1969).

In order to evaluate quantitatively the anti-DNA/RNA hybrid molecule immunolabelling, the labelling density (gold grains/ μ m²) over nucleoplasm and nucleolus was calculated on 20 micrographs ($\times 26,000$) from control and treated cells, by using AnalySIS Image processing (Soft Imaging System GmbH, Germany). For background evaluation, the resin outside the cells was considered.

All statistical comparisons were performed by the one-way ANOVA test; moreover, in order to determine which pairs of samples tended to differ, the standard procedure of multiple comparisons for the test was used. Statistical significance was set at $P \leq 0.05$.

3. Results

The exposure of cells to low concentrations of Roundup (1 and 10 mM) for 4–48 h does not increase cell mortality: the Trypan blue exclusion test showed that dead cells were always less than 2% of the total cells, without statistically significant difference with the controls.

The cytometric evaluation of the DNA content confirmed that the distribution of cells in the different cell cycle phases was similar in control and treated HTC cells, and, in particular, that the S-phase fraction did not change, ranging from 32% to 35% under all the conditions tested. This indicates that the exposure to Roundup at these concentrations does not affect the cytokinetic characteristics of this cell system.

Based on these results, for the following tests we exposed HTC cells to 10 mM Roundup for a short and a long time exposure, i.e., for 4 and 24 h. The morphometric analysis (Table 2) demonstrates that, compared to the untreated controls, Roundup-treated cells did not undergo significant changes in their nuclear and cytoplasmic areas, nor in the nucleus-to-cytoplasm ratios. However, an increase in the number and size of cytoplasmic vacuoles was apparent in treated cells (Fig. 1).

The immunolabelling experiments in fluorescence microscopy showed that, following exposure to Roundup, the microtubule and actin cytoskeleton, the Golgi apparatus, and mitochondria did not undergo evident changes in morphology, quantity and intracellular distribution (Fig. 2).

This was confirmed for mitochondria, on a quantitative basis, by the morphometric analysis of electron micrographs, which demonstrated that the total area did not change (Table 3); on the contrary, the inner-to-outer membrane ratio was significantly lower in Roundup-treated cells, suggesting that the mitochondrial function could be affected. Consistently, after Roundup treatment, there was a drop in the mitochondrial potential ($\Delta\psi$ m). This was measured using the fluorescent probe JC-1, which exhibits a potential-dependent accumulation in mitochondria. In untreated samples, three sub-populations can be observed (encircled in Fig. 3): one exhibits high red and relatively low green signals, and is for cells containing most energized mitochondria; a second one has similarly high red signals but increasing green fluorescence and represents a transition cell sub-population with lower mitochondrial membrane potential; the third, much less abundant, sub-population with low red and high green fluorescence contains cells with less energized mitochondria. After 4 and 24 h treatment with Roundup, the shift to higher green fluorescence signals of

Table 2Means $\pm$ SE values of variables considered in HTC cells

	Cellular area (μm^2)	Nuclear area (μm^2)	Cytoplasmic area (μm^2)	N/C ratio	Pore frequency (pore nr/ μm)	PG density (PG nr/ μm^2)	Nucleolar area (μm^2)	FC area (μm^2)	FC (%)	DFC (%)	GC (%)
Control	333.78 $\pm$ 12.83	85.71 $\pm$ 2.91	248.07 $\pm$ 10.99	0.38 $\pm$ 0.01	1.97 $\pm$ 0.22	1.63 $\pm$ 0.32*	3.22 $\pm$ 0.45	0.011 $\pm$ 0.005	0.06 $\pm$ 0.01	34.27 $\pm$ 2.21	63.70 $\pm$ 2.43
Roundup 10 mM (4 h)	349.65 $\pm$ 10.62	89.11 $\pm$ 3.12	252.77 $\pm$ 11.02	0.37 $\pm$ 0.01	1.79 $\pm$ 0.21	2.15 $\pm$ 0.13	3.89 $\pm$ 0.31	0.013 $\pm$ 0.002	0.05 $\pm$ 0.01	30.6 $\pm$ 2.83	66.19 $\pm$ 2.99
Roundup 10 mM (24 h)	326.22 $\pm$ 9.47	88.86 $\pm$ 3.40	237.35 $\pm$ 7.47	0.39 $\pm$ 0.02	2.15 $\pm$ 0.75	2.32 $\pm$ 0.18	2.83 $\pm$ 0.44	0.015 $\pm$ 0.001*	0.06 $\pm$ 0.01	27.97 $\pm$ 1.22	59.89 $\pm$ 1.51
	<i>P</i> = 0.69	<i>P</i> = 0.56	<i>P</i> = 0.50	<i>P</i> = 0.64	<i>P</i> = 0.42	<i>P</i> = 0.001	<i>P</i> = 0.22	<i>P</i> = 0.04	<i>P</i> = 0.31	<i>P</i> = 0.02	<i>P</i> = 0.19

In each column, values identified by asterisks are significantly different from the others.

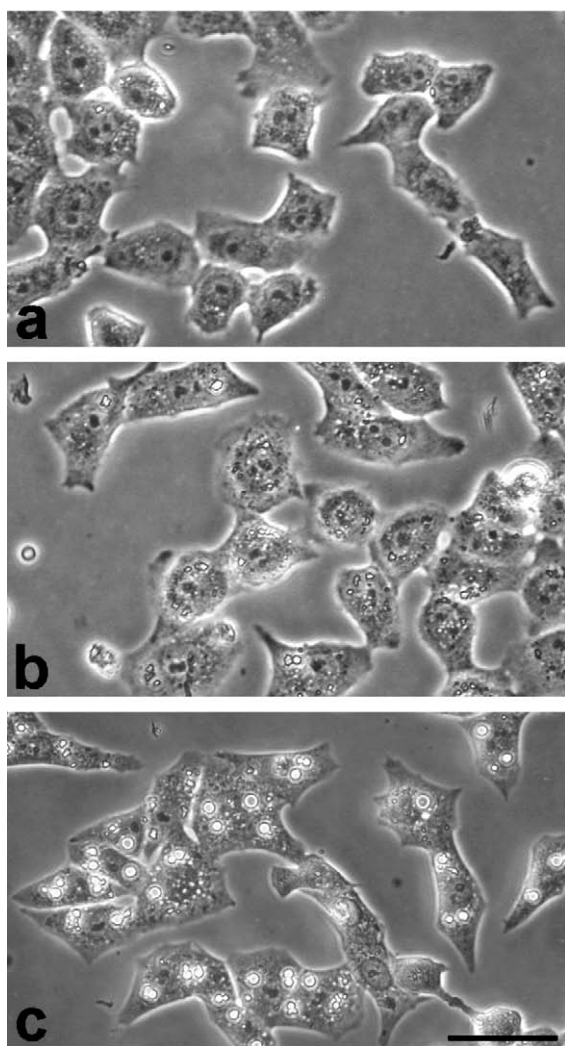


Fig. 1. Phase contrast micrographs of HTC cells: (a) untreated controls; (b) 1 mM Roundup-treated cells for 4 h and (c) 1 mM Roundup-treated cells for 24 h. In cells exposed to Roundup, a progressive increase of cytoplasmic vacuole size and number is observed, in the absence of major morphological changes. Bar = 50 μm .

most cells indicates the massive collapse of the mitochondrial potential (Fig. 3).

The lysosomes were also found to be affected in HTC cells treated with 10 mM Roundup for 24 h. They were apparently more numerous than in controls (Fig. 2) and showed an increased intracellular density (mean values $\pm$ SE: 6.42 $\pm$ 0.21 lysosomes/100 μm^2 vs. 5.01 $\pm$ 0.42 in controls; *P* = 0.001).

In electron microscopy (Fig. 4), HTC control cells showed numerous ribosomes occurring both free in the cytoplasm or associated with the endoplasmic reticulum, well developed Golgi complexes, some smooth endoplasmic reticulum, and mitochondria with lamellar transversal cristae. In the nuclei, condensed chromatin was quite scarce and mainly distributed at the nuclear periphery and in association with nucleoli. As usual, perichromatin fibrils and PGs were observed at the edge of condensed chromatin, while interchromatin granules were distributed in the nucleoplasm. In the nucleoli many FC surrounded by the DFC, and a prominent GC were found. In all Roundup-treated cells, free ribosomes, rough and smooth endoplasmic reticulum and Golgi complexes appeared to be unchanged. However, in parallel with the already mentioned decrease in the mitochondrial inner/outer membrane length ratio (Table 3), numerous vacuoles often containing cellular debris and residual bodies accumulated in the cytoplasm. At the nuclear level, the RNP-containing structures in the nucleoplasm and in the nucleolus did not show any apparent modification, although morphometrical analyses revealed an accumulation of PG both after 4 and 24 h of Roundup treatment as well as a significant increase of FC size and a decrease in DFC amount after a 24 h of Roundup treatment (Table 2).

As expected, the antibody recognizing DNA/RNA hybrid molecules (which may be considered as an *in situ* marker of transcription) specifically labelled perichromatin fibrils and the nucleolar DFC in all cell samples (Fig. 5). Significant quantitative differences were, however, observed between control and Roundup-treated cells (Table 4). In detail, both the nucleoplasmic and nucleolar anti-DNA/RNA hybrid labelling of control cells were significantly higher than those found in all Roundup-treated cells, while no significant difference was found between cells treated for 4 and 24 h (*P* = 0.067 for the nucleoplasm and *P* = 0.490 for the nucleolus).

4. Discussion

At the low concentration of Roundup used and for the exposure times considered, the death rate and the general morphological features of HTC cells were not affected. Namely, the size and shape of the whole cell, the cytoplasm, and the nucleus were not modified. The distribution of cells in the cycle, and in particular the fraction of S-phase cells, did not change significantly, thus suggesting that the presence of the herbicide in the culture medium did not affect the cytokinetic characteristics of this cell population. Most of the cytoplasmic organelles (i.e., the rough and smooth endoplasmic reticulum, the free ribosomes, the Golgi apparatus as well as the actin and tubulin cytoskeleton) did not change even for the longest treatment times.

Only the lysosomes and mitochondria were apparently affected. In HTC-treated cells, the lysosome density increased and, at the ultrastructural level, the organelles appeared to contain membrane

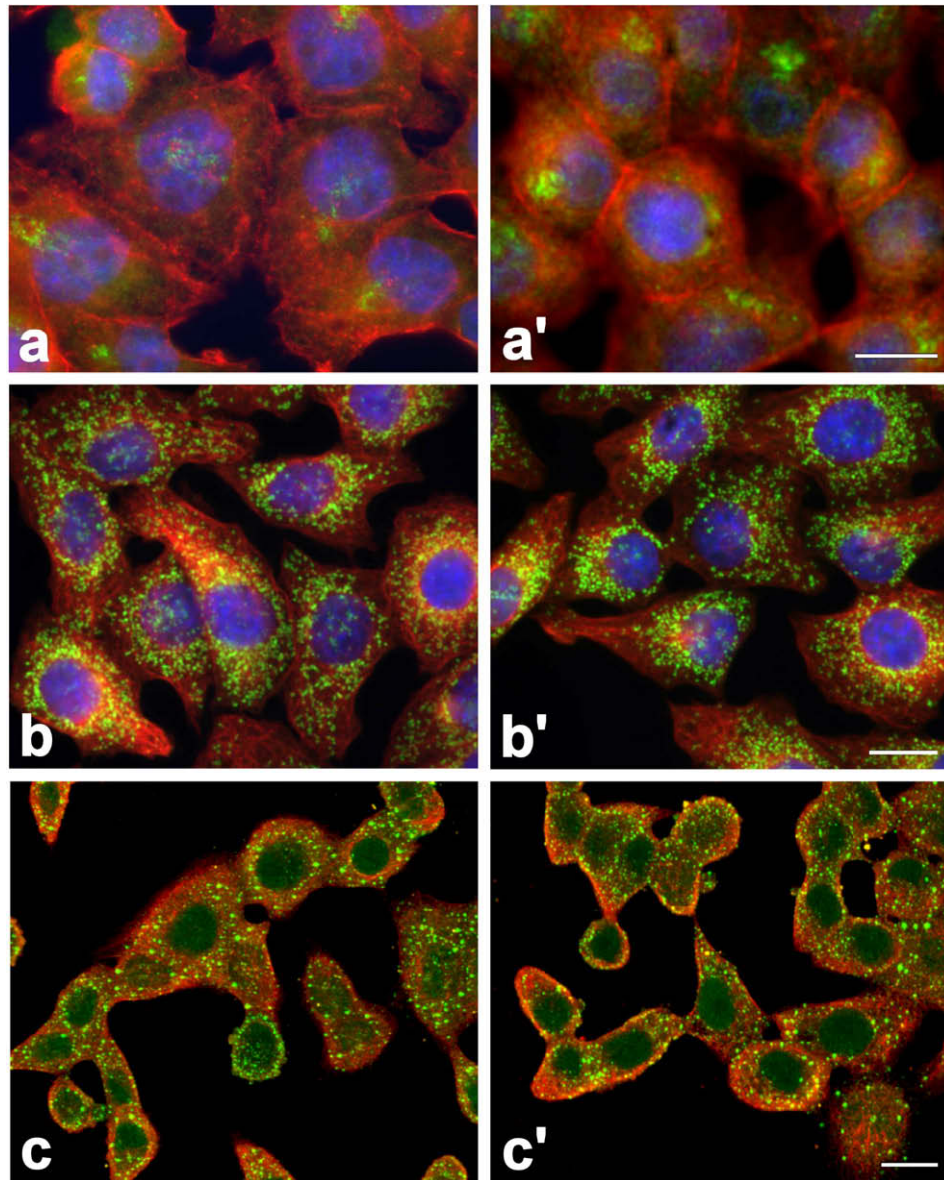


Fig. 2. Untreated controls (left column) and Roundup (1 mM)-treated HTC cells for 24h (right column). (a) and (a') Dual labelling for the Golgi apparatus (human autoimmune serum, Alexa 488-conjugated secondary antibody, green fluorescence) and actin (Alexa 594-conjugated phalloidin). No apparent changes were found; the blue fluorescence is for DNA after staining with Hoechst 33258. (b) and (b') Dual immunolabelling for mitochondria (human autoimmune serum, Alexa 488-conjugated secondary antibody, green fluorescence) and α -tubulin (Alexa 594-conjugated antibody, red fluorescence); Hoechst 33258 blue fluorescence for DNA. (c) and (c') Immunolabelling with a human autoimmune serum recognizing a lysosomal protein finally labelled with an Alexa 488-conjugated antibody (green fluorescence). Lysosomes are more numerous in Roundup-treated cells than in controls (see also text for the quantitative results); cytoplasm counterstaining with Trypan blue (red fluorescence). (a) and (b) conventional fluorescence microscopy and (c) confocal microscopy. Bars = 20 μ m. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

Table 3

Means $\pm$ SE values of variables considered in mitochondria of HTC cells

	Mitochondrial area (μ m ²)	Inner/outer membrane ratio
Control	0.21 $\pm$ 0.01	1.90 $\pm$ 0.08*
Roundup 10 mM (4 h)	0.24 $\pm$ 0.03	1.21 $\pm$ 0.06
Roundup 10 mM (24 h)	0.21 $\pm$ 0.01	1.24 $\pm$ 0.08
	$P = 0.70$	$P < 0.001$

In each column, values identified by asterisks are significantly different from the others.

and cytosolic debris. This suggests that cytoplasmic damages are induced by the presence of Roundup, with the subsequent increase of the lysosome activity to remove injured structures.

The results of the analyses by flow cytometry, fluorescence and electron microscopy revealed that Roundup treatment induces morpho-functional modifications also in the mitochondria, which underwent membrane depolarization and a reduction of the inner membrane length indicating a decline in respiratory activity; the amount and intracellular distribution of mitochondria remained, however, unaltered. Accordingly, high concentration of Roundup has been reported to alter the mitochondrial inner membrane (Szarek et al., 2000) and to lower the cell respiratory activity (Peixoto, 2005). It is interesting to note that in hepatocyte mitochondria of old mice fed from their weaning the glyphosate-tolerant GM soybean a decreased expression of some respiratory enzymes and a reduction of the cristae were also observed (Malatesta et al., 2008).

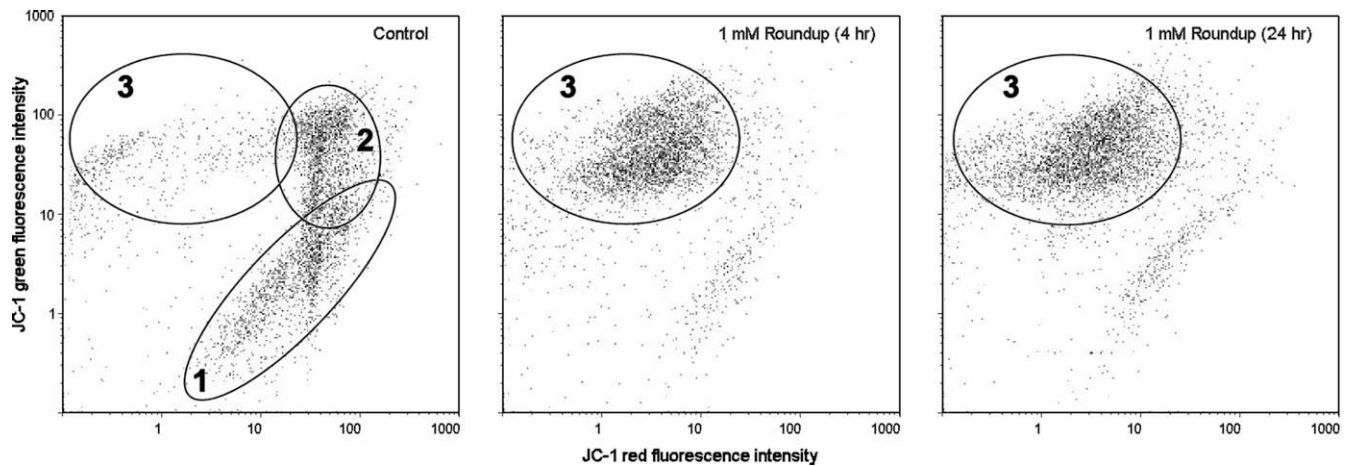


Fig. 3. Dual-parameter cytograms of untreated controls (left panel), and Roundup-treated HTC cells (1 mM for 4 and 24 h: middle and right panel, respectively) after vital staining with JC-1. In abscissa, red fluorescence signals; in ordinate, green fluorescence signals. In untreated samples, three sub-populations can be observed: one exhibits high red and relatively low green signals (encircled area 1), a second one shows high red signals and increasing green fluorescence (encircled area 2), the third one shows low red and high green fluorescence (encircled area 3). After 4 and 24 h treatment with Roundup, most cells shift to higher green fluorescence signals (encircled area 3).

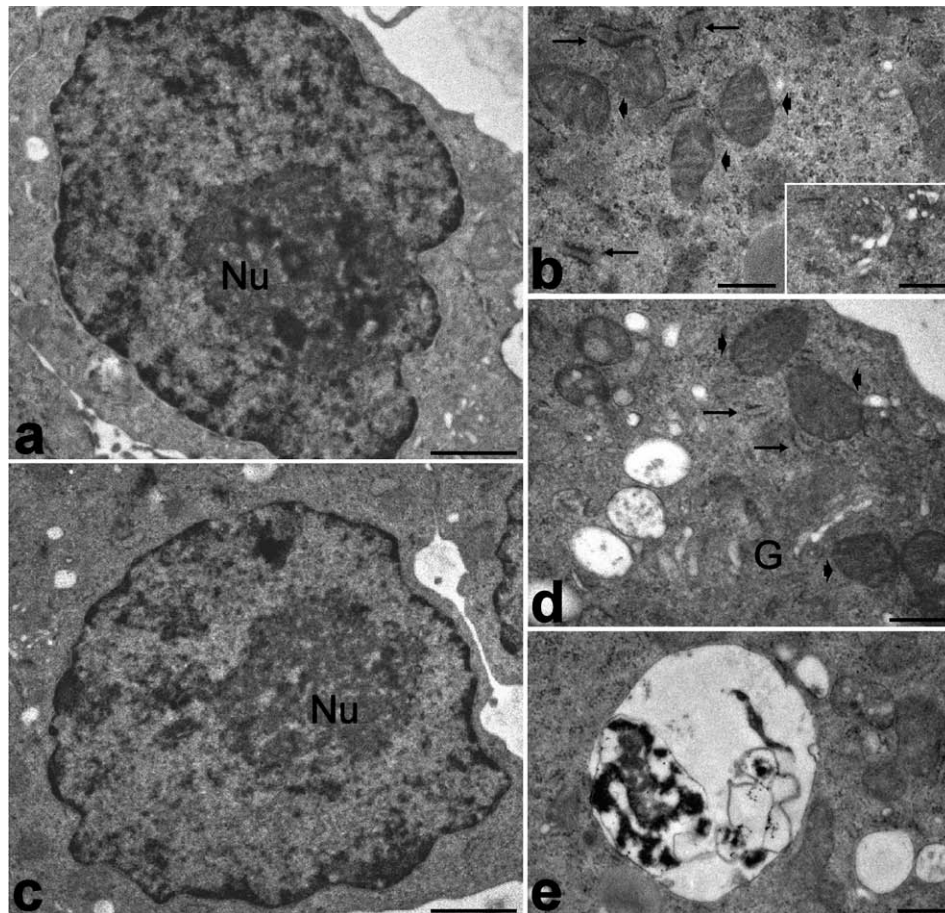


Fig. 4. Electron micrographs of control (a) and (b) and 10 mM Roundup-treated (c)–(e) HTC cells; Epon embedding, uranyl acetate and lead citrate staining. Control cells (a) and (b) show nuclei (a) with scarce condensed chromatin and nucleoli (Nu) with evident FC and DFC; in the cytoplasm (b) numerous ribosomes occur both free and associated with the endoplasmic reticulum (arrows), Golgi complexes are well developed (inset) and mitochondria show lamellar transversal cristae (arrowheads). Roundup-treated cells (c)–(e) differ from control cells because of nucleoli (Nu) with scarce DFC (c), mitochondria (arrowheads) with less developed inner membrane (d), and many vacuoles and residual bodies accumulating in the cytoplasm (d). Some vacuoles have been found to contain cellular debris (e). Conversely, rough endoplasmic reticulum (arrows) and Golgi complex (G) are not modified by Roundup treatment (d). Bars = 1 μ m (a) and (c) or 0.5 μ m (b), (d) and (e).

In the nuclei of Roundup-treated HTC cells, the general appearance of the RNP-containing structural constituents was similar as

in controls, with the exception of the PG, which accumulated in the interchromatin space. These RNP structures usually act in the

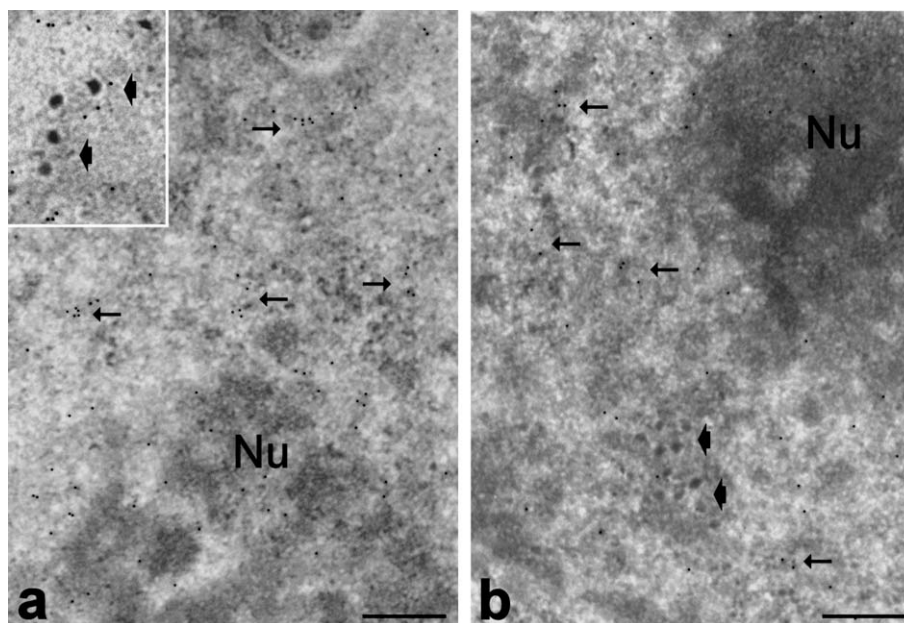


Fig. 5. Electron micrographs of control (a) and 10 mM Roundup-treated (b) HTC cells; LRWhite embedding, immunolabelling with anti-DNA/RNA hybrid molecule antibody, EDTA staining. Labelling occurs both in the nucleoplasm, mainly associated to PF (arrows) and in the nucleolus (Nu). PG (arrowheads), which accumulate in Roundup-treated cells (b), are not labelled. Note the weaker signal on the Roundup-treated cell. The gold grain contrast was digitally enhanced on micrographs by using Adobe Photoshop. Bars = 0.2 μm .

Table 4

Mean values $\pm$ SE of labelling densities (gold grains/ μm^2) obtained with anti-DNA/RNA hybrid molecule antibody on control and Roundup-treated HTC cells

	Nucleoplasm	Nucleolus
Control	20.74 $\pm$ 0.67*	19.59 $\pm$ 1.79*
Roundup 10 mM (4 h)	15.67 $\pm$ 0.91	12.77 $\pm$ 0.87
Roundup 10 mM (24 h)	13.46 $\pm$ 0.73	11.38 $\pm$ 0.97
	$P < 0.001$	$P = 0.010$

In each column, values identified by asterisks are significantly different from the others. Background levels were negligible (0.18 $\pm$ 0.09 gold grains/ μm^2).

nucleus as both vectors and storage sites for already spliced pre-mRNA (Fakan, 2004), and they were found to accumulate in different experimental systems as a consequence of altered pre-mRNA processing or of an impaired intranuclear or nucleus-to-cytoplasm transport of mature mRNAs (see, for instance, Puvion-Dutilleul and Puvion, 1981; Lafarga et al., 1993; Zancanaro et al., 1993). Interestingly again, pancreatic acinar cells and spermatocytes of young and adult mice fed GM soybean (Malatesta et al., 2003; Vecchio et al., 2004), as well as the hepatocytes of old mice fed the same diet showed higher amounts of PG than the mice fed non-GM soybean (Malatesta et al., 2008).

In addition to PG accumulation, in Roundup-treated cell nuclei, an increase in size of the nucleolar FCs took place in parallel with a decrease in the amount of the DFC. The nucleolus is a very dynamic structure whose activity and architecture are modulated by changes in the cell metabolic conditions (Schwarzacher and Wachtler, 1993; Shaw and Jordan, 1995). In particular, when the metabolic rate decreases the number of large FCs increases and the DFC shrinks (e.g. Jordan and McGovern, 1981; Schwarzacher and Wachtler, 1993).

The described evidence suggests that Roundup-treated HTC cells may undergo a reduction in their nuclear activity. This is consistent with the observed lower amounts of DNA/RNA hybrid molecules in Roundup-treated cells, which strongly points to a reduced transcriptional activity (Testillano et al., 1994). Although no direct evidence exists for an effect of glyphosate on transcription and

translation, it has been shown that Roundup, when used at high concentrations, generates DNA adducts (Clements et al., 1997; Peluso et al., 1998) and determines cell division dysfunction and transcription inhibition (Marc et al., 2002, 2005). A reduction in transcription and splicing activity has been reported also in pancreatic acinar cells and spermatocytes of young and adult mice fed GM soybean (Malatesta et al., 2003; Vecchio et al., 2004), as well as in hepatocytes of old mice fed the same diet (Malatesta et al., 2008).

Taken together, the data obtained on HTC cells demonstrate that low concentrations of Roundup do not affect the whole structural characteristics of cellular components, revealing that absence of cellular alterations does not necessarily mean absence of exposure to the herbicide. It is noteworthy that the effects are quite similar after short or long exposures, suggesting that cells are able to respond to the herbicide toxicity by establishing a new steady state at a lower metabolic rate and by eliminating the potentially dangerous damaged cellular components via lysosomes. Consequently, viability and cell cycle are not affected by treatment with low concentrations of Roundup.

When the *in vitro* effects of low concentrations of Roundup are compared with the modifications induced by the intake of a glyphosate-tolerant GM soybean in mouse tissues, many similarities become evident. In particular, most mouse tissues analysed show signs of metabolic lowering (Malatesta et al., 2002b, 2003; Vecchio et al., 2004), with the exception of hepatocytes in young animals (Malatesta et al., 2002a). In this case, however, the hepatic role in degrading and detoxifying toxic compounds received from the intestines or the general circulation could contribute to maintain hepatocyte activity at a high level; this prolonged activation probably induces a more marked age-related metabolic depression in old GM-fed mice (Malatesta et al., 2008).

We cannot exclude that other factors than the presence of the herbicide residues could be responsible for the cellular modifications described in GM-fed mice. For instance, GM soybean has been reported to contain amounts of phytoestrogens that are lower than the non-GM ones (Lappé et al., 1998) and, taking into account the influence these compounds exert at multiple levels of cell activity

(Cassidy, 2003; Plessow et al., 2003), a difference in their daily intake could play a role in the tissue modifications in mice fed GM soybean.

However, there is an interesting concordance between the morpho-functional modifications found in HTC cells exposed to low concentrations of Roundup and those observed in tissues of mice fed a GM soybean-based diet: this would support the hypothesis that Roundup residues in the crop may be involved in the cytological changes *in vivo*. Although there is no evidence for a direct effect of Roundup in inducing major cell damage or metabolic defects, we cannot exclude that a long-lasting intake, through the diet, of herbicide residues might interfere with different metabolic pathways.

Conflict of Interest Statement

The authors declare that there are no conflicts of interest.

Acknowledgments

Thanks are due to Mrs. Paola Veneroni and Dr. Beatrice Baldelli for their excellent technical assistance. The human autoimmune sera recognizing proteins of the Golgi apparatus, the lysosomes, and the mitochondria were a kind gift of Dr. Claudia Alpini, IRCCS San Matteo, Pavia, Italy. This work was supported by the Agenzia Servizi Settore Agroalimentare delle Marche, Italy: this sponsor did not have any involvement in the study processing nor in the decision to submit the paper for publication.

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